

4 More of English: Nonverbal Predicates, Modifiers, Definite Descriptions

We ultimately want to determine appropriate types of meanings for lexical items of all syntactic categories and to predict how these meanings are composed in all sorts of syntactic constructions. In this chapter, we take some further small steps towards this goal. We continue with the Fregean working hypothesis that there are two types of basic, or saturated, meanings – namely, individuals and truth-values – and that all other meanings are functions that are somehow constructed out of these.

We will run into important general issues like vagueness, context-dependency, and presupposition, but will not stop to consider them seriously and systematically. We will also make quite a few concrete decisions without defending them against alternatives. The primary purpose of this chapter is to build a reasonably broad base of simple constructions for which we have at least a preliminary treatment, so that we are not too constrained in our choice of examples in the subsequent chapters. A secondary purpose is to give an overview of some of the basic issues regarding modification and presupposition.

4.1 Semantically vacuous words

Some lexical items are widely held to make no semantic contribution to the structures in which they occur. The standard example is of certain occurrences of prepositions, such as “of” in “proud of John” or “father of John”. Another plausible candidate is the copula “be” in predicative sentences like “John is rich”. We will also assume, at least for the time being, that the indefinite article “a” is vacuous when it occurs in predicate nominals such as “a cat” in “Kaline is a cat”.¹ We would want the following equalities, for example:

- (1) $\llbracket \text{of John} \rrbracket = \llbracket \text{John} \rrbracket$
 $\llbracket \text{be rich} \rrbracket = \llbracket \text{rich} \rrbracket$
 $\llbracket \text{a cat} \rrbracket = \llbracket \text{cat} \rrbracket$

There are various ways of making this come out. One way is to list semantically vacuous items in the lexicon as denoting the identity function of the appropriate type, for instance:

- (2) $\llbracket \text{of} \rrbracket = \lambda x \in D_e . x$
 (3) $\llbracket \text{be} \rrbracket = \lambda f \in D_{\langle e, t \rangle} . f$
 (4) $\llbracket \text{a} \rrbracket = \lambda f \in D_{\langle e, t \rangle} . f$

In words: $\llbracket \text{of} \rrbracket$ is that function which maps every individual in D_e to itself, and $\llbracket \text{be} \rrbracket$ ($= \llbracket \text{a} \rrbracket$) is that function which maps every function in $D_{\langle e, t \rangle}$ to itself.

An even easier possibility is to assume that the semantic component simply “doesn’t see” such items. In other words, a structure that is really binary-branching may be treated as non-branching in the semantics: a branch occupied only by a vacuous item doesn’t count. The principle for nonbranching nodes then applies and passes up the meaning unchanged. Either way, we ensure the equalities in (1).

4.2 Nonverbal predicates

What we have assumed for verbs can be extended straightforwardly to adjectives, nouns, and prepositions. Just as intransitive verbs denote functions from individuals to truth-values, so do many nouns and adjectives, for instance:

- (1) $\llbracket \text{cat} \rrbracket = \lambda x \in D_e . x \text{ is a cat}$
 (2) $\llbracket \text{gray} \rrbracket = \lambda x \in D_e . x \text{ is gray}$

Among prepositions, intransitive (1-place, monadic) ones are the exception rather than the rule, but there are some candidates:

- (3) $\llbracket \text{out} \rrbracket = \lambda x \in D_e . x \text{ is not in } x\text{'s home}$

Each of these categories also has transitive (2-place, dyadic) members, whose extensions are just like those of transitive verbs – for example, “part” in “part of Europe” and “proud” in “proud of John” (note the vacuity of “of”), and all run-of-the-mill prepositions:

(4) $\llbracket \text{part} \rrbracket = \lambda x \in D_e . [\lambda y \in D_e . y \text{ is part of } x]$

(5) $\llbracket \text{fond} \rrbracket = \lambda x \in D_e . [\lambda y \in D_e . y \text{ is fond of } x]$

(6) $\llbracket \text{in} \rrbracket = \lambda x \in D_e . [\lambda y \in D_e . y \text{ is in } x]$

These lexical entries allow us to calculate appropriate denotations for the phrases “part of Europe”, “fond of Joe”, “in Texas”, by means of our composition principle Functional Application (FA), for instance:

By FA: $\llbracket \text{in Texas} \rrbracket = \llbracket \text{in} \rrbracket(\llbracket \text{Texas} \rrbracket)$

By lexical entry for Texas: $\llbracket \text{Texas} \rrbracket = \text{Texas}$

Hence: $\llbracket \text{in Texas} \rrbracket = \llbracket \text{in} \rrbracket(\text{Texas})$

By lexical entry for in: $\llbracket \text{in} \rrbracket(\text{Texas})$

$= [\lambda x \in D_e . [\lambda y \in D_e . y \text{ is in } x]](\text{Texas}) = \lambda y \in D_e . y \text{ is in Texas.}$

Hence: $\llbracket \text{in Texas} \rrbracket = \lambda y \in D_e . y \text{ is in Texas.}$

We will disregard the case of ditransitive (3-place, triadic) predicates, though there are presumably some analogs to verbs like “give” and “introduce” in other syntactic categories.

Exercise

Calculate the truth-conditions for at least one of the sentences “Joe is in Texas”, “Joe is fond of Kaline”, and “Kaline is a cat”.

4.3 Predicates as restrictive modifiers

It has often been observed that prepositional phrases (PPs) may appear inside NPs in three distinct semantic roles: as arguments, as restrictive modifiers, or as nonrestrictive modifiers.² Typical examples of each type are the following:

- (1) a part *of Europe* (argument)
- (2) a city *in Texas* (restrictive modifier)
- (3) Susan, *from Nebraska*, (nonrestrictive modifier)

We have already dealt with the first case: PPs that are arguments are headed by vacuous prepositions, thus they have the same denotations as the NPs they contain (individuals), and they are arguments of 2-place (relational) nouns such as “part”.

About the third case, we have next to nothing to say, and we mentioned it only to guard against confusing nonrestrictive modifiers with restrictive ones. The basic intuition that most authors have expressed about the semantics of nonrestrictive modification³ is that nonrestrictive modifiers are not semantically composed at all with the phrases they modify. Rather, they have the status of separate sentences which serve to make side-remarks of some kind. For example, the meaning of (4) is not unlike that of (5).

- (4) It is surprising that Susan, from Nebraska, finds it cold in here.
- (5) It is surprising that Susan finds it cold in here. Note that she is from Nebraska.

This makes it reasonable to assume that at the level at which our semantic rules apply, the nonrestrictive modifier isn’t part of the structure at all, so the question of how its denotation should be composed with that of its modifier doesn’t arise in the first place. This said, we will concentrate on restrictive modifiers, which are our primary topic in this section.

While the distinction between arguments and restrictive modifiers is notoriously difficult to make in practice, the basic semantic intuition behind it is simple: Arguments reduce the adicity of the noun they combine with; modifiers leave it unchanged.

What it means to reduce the adicity of the noun is illustrated by the example we have treated: “part” is a 2-place predicate, while the result of combining it with “of Europe” to form “part of Europe” is a 1-place predicate. Restrictive modifiers, by contrast, are characterized by the fact that they leave the semantic type, including the adicity, of the modifier completely unchanged. Intuitively, “city in Texas” has the same kind of extension as “city”: namely, the characteristic function of a set. More specifically, if $\llbracket \text{city in Texas} \rrbracket$ is the characteristic function of a set A and $\llbracket \text{city} \rrbracket$ is the characteristic function of a set B, then A is a subset of B: namely, that subset which results by intersecting B with the set of things in Texas. Consider the truth-conditions of “Lockhart is a city in Texas” to confirm this intuition.

Now we have already proposed an analysis of “in” above (motivated at the time by occurrences of this preposition in simple copula + PP phrases), under which “in Texas” has a denotation of type $\langle e, t \rangle$. Given this decision and our current inventory of composition rules, we predict “city in Texas” to be uninterpretable. As a branching structure with daughters “city” and “in Texas”, we should be interpreting it by applying either $\llbracket \text{city} \rrbracket$ to $\llbracket \text{in Texas} \rrbracket$ or vice versa. But neither is possible, as both are functions of type $\langle e, t \rangle$.

What shall we do about this problem? There are two directions we could take: either revise out lexical semantics for (some of) the ingredients, or else stipulate a new composition rule. We will entertain both options, beginning with the second.

4.3.1 A new composition rule

Here is a composition principle which is tailored to the situation at hand.

(6) *Predicate Modification* (PM)⁴

If α is a branching node, $\{\beta, \gamma\}$ is the set of α 's daughters, and $\llbracket \beta \rrbracket$ and $\llbracket \gamma \rrbracket$ are both in $D_{\langle e, t \rangle}$, then

$$\llbracket \alpha \rrbracket = \lambda x \in D_e . \llbracket \beta \rrbracket(x) = \llbracket \gamma \rrbracket(x) = 1.^5$$

Applied to “city in Texas”, (6) gives the desired result:

(7) $\llbracket \text{city in Texas} \rrbracket$

$$= \lambda x \in D_e . \llbracket \text{city} \rrbracket(x) = \llbracket \text{in Texas} \rrbracket(x) = 1$$

$$= \lambda x \in D_e . x \text{ is a city and } x \text{ is in Texas.}$$

(In the last step, we used the lexical entries and the result of a previous calculation from section 4.2 above.)

PM is general enough to cover not just PPs modifying nouns, but likewise adjective phrases (APs), whether to the right or the left of a noun, and also stacked modifiers in unlimited numbers. For instance, we can now predict correctly the truth-conditions of (8).

(8) Kaline is a gray cat in Texas fond of Joe.

To treat this example, we must, of course, impose some binary-branching hierarchy among the three modifiers “gray”, “in Texas”, and “fond of Joe”. Given the nature of the semantic operation performed by PM, it so happens that all our different choices in this regard yield logically equivalent results. This is as it should be. The syntax of English (to the best of our knowledge) does not determine a unique parse for this sentence, but it is nevertheless perceived as truth-conditionally unambiguous.

Exercise

Calculate the truth-conditions for (8), given one possible syntactic parse.

The operation performed by PM has also been called “intersective modification”, because if we look at the sets instead of at their characteristic functions, it amounts to set-theoretic intersection. “Conjunctive composition” would be another natural name, highlighting the connection with the semantics of “and”.⁶ (Notice that “city in Texas” receives exactly the meaning we derive for “city *and* in Texas”, using the predicate-coordinating “and” we defined in section 3.4.)

Predicate modification is a genuinely new semantic composition principle on our list. It is obviously not functional application. If it is really needed, there is more to semantic composition than Frege’s Conjecture. Are we forced to this conclusion?

4.3.2 Modification as functional application

As we have already mentioned, the alternative to PM is to explore revised lexical entries for the words that may head modifiers. Suppose we insist that $\llbracket \text{city} \rrbracket$ and $\llbracket \text{in Texas} \rrbracket$ combine by Functional Application after all. If we keep $\llbracket \text{city} \rrbracket$ as before, with type $\langle e, t \rangle$, then $\llbracket \text{in Texas} \rrbracket$ will have to be of type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$. This in turn requires a new semantics for at least one of “in” and “Texas”. If we keep the assumption that $\llbracket \text{Texas} \rrbracket = \text{Texas} \in D_e$, we must reinterpret “in”. It must now denote a function of type $\langle e, \langle \langle e, t \rangle, \langle e, t \rangle \rangle \rangle$. Which such function? Well, we know that we want to be able to derive equations like these:

- (9) $\llbracket \text{in} \rrbracket(\text{Texas})(\lambda x \in D_e . x \text{ is a city})$
 $= \lambda x \in D_e . x \text{ is a city and } x \text{ is in Texas.}$
 $\llbracket \text{in} \rrbracket(\text{Building 20})(\lambda x \in D_e . x \text{ is a room})$
 $= \lambda x \in D_e . x \text{ is a room and } x \text{ is in Building 20.}$

The generalization appears to be that, for any individual $y \in D_e$ and any function $f \in D_{\langle e, t \rangle}$, $\llbracket \text{in} \rrbracket(y)(f) = \lambda x \in D_e . f(x) = 1$ and x is in y . This determines directly the desired definition for the function $\llbracket \text{in} \rrbracket$:

- (10) $\llbracket \text{in} \rrbracket = \lambda y \in D_e . [\lambda f \in D_{\langle e, t \rangle} . [\lambda x \in D_e . f(x) = 1 \text{ and } x \text{ is in } y]]$

By similar reasoning, we can determine an entry for the adjective “gray” that permits phrases like “gray cat” to be interpreted by Functional Application:

- (11) $\llbracket \text{gray} \rrbracket = \lambda f \in D_{\langle e, t \rangle} . [\lambda x \in D_e . f(x) = 1 \text{ and } x \text{ is gray}]$

Exercise

Calculate the truth-conditions for (8), given one possible syntactic parse. This time, use FA instead of PM.

By systematically revising the entries of all adjectives and prepositions, we are able to interpret all phrases containing a noun with one or more modifiers in them by means of Functional Application alone, and so we can eliminate Predicate Modification from the theory.

But there is a trade-off. What happens now when an AP or PP stands as a (maximal) predicate by itself, as in “Julius is gray” or “Julius is in Amherst”? If APs and PPs denote functions of type $\langle\langle e,t\rangle,\langle e,t\rangle\rangle$, these sentences are *prima facie* uninterpretable. We could try to solve the problem by assigning a suitable denotation to the copula “be”.

Exercise

Define such a denotation. There are two distinct solutions.

But then this same “be” would not be interpretable in sentences with a nominal predicate, for example, “Julius is a cat”, since NPs like “a cat” are still of type $\langle e,t\rangle$. So the copula would have to be ambiguous between vacuous and nonvacuous occurrences.

Another solution is to assume a systematic lexical ambiguity in all adjectives and prepositions. Each has both the initially assumed type $\langle e,t\rangle$ (or $\langle e,\langle e,t\rangle\rangle$) meaning and the new type $\langle\langle e,t\rangle,\langle e,t\rangle\rangle$ (or $\langle e,\langle\langle e,t\rangle,\langle e,t\rangle\rangle\rangle$) denotation. The syntax may freely generate both homonyms in all the same places, but the Principle of Interpretability will allow only one in any given environment.

Exercise

It would not be adequate to list the two readings of each preposition or adjective separately in the lexicon, as if they had to be learned individually. Evidently, there is a systematic relation between the two readings, which makes one predicable, given the other. So we would want to list only one in the

lexicon, and derive the other by means of a general "lexical rule". Spell out two versions of this proposal. For the first version, assume that the homonyms with the simpler types ($\langle e, t \rangle$ or $\langle e, \langle e, t \rangle \rangle$) are basic and listed in the lexicon. For the second version, assume that the more complicated types $\langle \langle e, t \rangle, \langle et \rangle \rangle$ or $\langle e, \langle \langle e, t \rangle, \langle et \rangle \rangle \rangle$ are the ones of the basic, individually listed items. Your task is to formulate the appropriate lexical rules for either version. That is, you have to specify general recipes that map arbitrary denotations of the basic type to secondary denotations of the appropriate nonbasic type. Rules of this kind are called "type-shifting rules".⁷

Yet another solution is to posit a certain amount of non-overt structure in VPs of the surface form "be" + AP or "be" + PP. Perhaps these VPs contain an invisible predicate that the AP or PP modifies, something like a zero equivalent of a bland noun like "thing" or "individual".

We will not explore or try to evaluate these options further. All we wanted to show here is that the elimination of the Predicate Modification rule is not without its price. As matters stand, it does not look entirely unreasonable if we decide to adopt PM after all.

4.3.3 *Evidence from nonintersective adjectives?*

Both analyses of adjectives that we have entertained so far predict that the following pair of sentences are logically equivalent:

- (12) (a) Julius is a gray cat.
 (b) Julius is gray and Julius is a cat.

In the analysis that uses PM, the equivalence follows directly from the content of the rule. We can prove it without using any specific information about the meanings of the lexical items, except the information about their semantic type. In the analysis that relies on Functional Application alone, the equivalence follows from the lexical meaning of the adjective "gray". On this alternative, mere inspection of the *types* of the words and the applicable composition principles does *not* suffice to prove it.

Since the equivalence in (12) is indeed intuitively valid, both analyses make the correct prediction, albeit in different ways. But it has often been noted⁸ that analogous equivalences do not obtain for many other adjectives, and it seems that this fact might have some bearing on the choice we have been contemplating.

Consider adjectives like "large" and "small". One might truthfully assert that a small elephant is still a very large animal. So it is intuitively possible for (13) to be true while (14) is false.

(13) Jumbo is a small elephant.

(14) Jumbo is a small animal.

This shows that (13) does not intuitively entail (14). But “Jumbo is an elephant” does entail “Jumbo is an animal”, and given the meaning of “and”, this implies that (15) entails (16).

(15) Jumbo is small and Jumbo is an elephant.

(16) Jumbo is small and Jumbo is an animal.

So (13) and (14) cannot be equivalent to (15) and (16) respectively. At least one, perhaps both, of these equivalences must be denied, or else we falsely predict that (13) entails (14).

Now we have seen that if adjectives are of type $\langle e, t \rangle$ and combine with their modifyees by PM, the equivalences in question follow regardless of specific lexical meaning. We have to conclude, therefore, that it is not possible to define an adequate type $\langle e, t \rangle$ meaning for “small”.

But if the meaning of “small” is of type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$, it seems that we have a chance of defining it appropriately, so that the inference from (13) to (14) does not go through. We do *not* want a lexical entry analogous to (11) above, of course (that is, we don’t want to define $\llbracket \text{small} \rrbracket$ as $\lambda f \in D_{\langle \langle e, t \rangle \rangle} . [\lambda x . f(x) = 1 \text{ and } x \text{ is small}]$), but fortunately there are other functions of type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$ to consider. Here is a proposal that seems to reflect the relevant intuition about why small elephants are not small animals.

(17) $\llbracket \text{small} \rrbracket =$
 $\lambda f \in D_{\langle \langle e, t \rangle \rangle} . [\lambda x \in D_e . f(x) = 1 \text{ and the size of } x \text{ is below the average size of the elements of } \{y : f(y) = 1\}]$

One might quibble with some of the details, but (17) is definitely on the right track towards an explanation of why (13) fails to imply (14). According to (17), (13) asserts that Jumbo is an elephant and Jumbo’s size is below the average elephant size. The set of all animals contains mostly individuals that are smaller than any elephants, so the average animal size is much lower than the average elephant size, and it is easy for Jumbo’s size to fall between the two. This is how (14) can be false when (13) is true.

So it seems that we should allow at least some adjectives to denote functions of type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$. “Small”, we have seen, cannot possibly be interpreted with the lower type $\langle e, t \rangle$, and this is just one of many examples for which the same kind of argument can be made. These adjectives are often called

nonintersective. In the adjective inventories of natural languages, they are apparently in the majority. “Intersective” adjectives – that is, those which validate equivalences like (12), like our initial example “gray” – represent the exception rather than the rule.

Are there any implications of this discussion for the analysis of the intersective adjectives? Not as far as we have seen up to now. Adding type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$ adjectives like “small” to the lexicon does not seem to cost us anything extra, whether we add them to a grammar that also has type $\langle e, t \rangle$ adjectives and a rule of PM, or to one where *all* adjectives have type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$ and PM is absent. Notice that our theory does not place any premium *per se* on having a uniform semantic type for all members of a given syntactic category. So the fact that on the PM option, some adjectives have type $\langle e, t \rangle$ and others type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$ is not in itself a reason to disprefer it.

The picture changes if we recall some of the trade-offs we thought to be tied to the elimination of PM. For instance, we observed that type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$ meanings are less straightforwardly interpretable than type $\langle e, t \rangle$ meanings in predicative occurrences of APs, as in “Julius is gray”. But it turns out that adjectives like “small”, which we have seen *cannot* have lexical entries of type $\langle e, t \rangle$, also occur as seemingly complete predicates:

(18) Jumbo is small.

What do we make of this? We seem to need one of the mechanisms that we considered in section 4.3.2 as a replacement for PM after all, whether we have PM or not. For instance, we may posit a zero modifiee in the syntax of (18), or assume a lexical rule that routinely produces secondary type $\langle e, t \rangle$ meanings from type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$ inputs. (See exercise above.) But this tips the balance, and a theory without PM begins to look more parsimonious over all.

Or does it? Let’s consider the whole case based on “small” a bit more carefully before we jump to conclusions. Is a type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$ analysis as in the entry (17) really forced on us by the data we have considered?

An elementary observation about adjectives like “small” that we have so far left aside is that they are vague and heavily context-dependent. In suitable specific discourse settings, people may have truth-value judgments about utterances containing “small” which are firm and uniform across speakers. But if we try to generalize over all felicitous uses of a given “small” sentence, we find that objects of practically any size can count as “small” and also as “not small”. It is important to note that this vagueness and context-dependency are not limited to predicative occurrences like the one in (18). It remains even when there is a noun that “small” modifies. Consider Jumbo again. When you first read sentence (13), which described him as “a small elephant”, in the context in which we presented it above, you spontaneously interpreted it in the manner we have

described: namely, as true only if Jumbo is smaller than the average elephant. But imagine we had first introduced a scenario populated with an army of monsters like King Kong. We might then have said something like: "Jumbo doesn't have a chance; he's only a small elephant", and this could have been true even if Jumbo were as large as or even larger than most other elephants.

So the meaning for "small" we codified in (17) represents at best a sort of default that applies when phrases of the form "small N" are interpreted more or less out of context. The contribution of the modified noun's meaning to the meaning of the whole phrase is not in general this mechanical. Perhaps the basic generalization about "small" is that it means "of a size less than the contextually salient standard". How the contextually salient standard is established for each given utterance of the word "small" is a complex affair. Previous discourse and the nonlinguistic circumstances of the utterance play a role. The mention of the word "elephant" in the immediate vicinity of the adjective draws attention to the elephant stereotype, including the stereotypical elephant size. Perhaps this is the whole reason why the average size of elephants happens to be the most salient standard in most situations where somebody utters the phrase "small elephant". In other words, the contribution of the modified noun may be rather indirect and mediated by the context.

If this picture is correct, a type $\langle e, t \rangle$ interpretation for "small" may be viable after all. The lexical entry might say essentially the following:

- (19) $\llbracket \text{small} \rrbracket = \lambda x \in D_e . x\text{'s size is below } c$, where c is the size standard made salient by the utterance context.

"Small elephant", as it occurs in (13), could then be interpreted by PM after all. It would receive the truth-conditions in (17) just in case the context of utterance does not supply a more salient size standard than the average size of elephants. To explain the intuition that (13) does not entail (14), we would assume that utterance contexts change quickly. We must acknowledge that it is possible, in fact highly natural, for an utterance of (14) which follows right after (13) to change the prevailing size standard from average elephant size to average animal size. We do predict that (13) entails (14) *if the context for both is the same*. But this prediction may be compatible with the evidence, if we can tell a plausible story about why the context will automatically change whenever an utterance of (14) follows one of (13).

We have evidently scratched only the surface of a complex of important issues here. Vagueness and context-dependency have been studied quite carefully by philosophers and linguists within the general framework of natural language semantics that we are presenting here. But the results of their work, both substantive and technical, are largely beyond the scope of this text.⁹ Our present conclusion must therefore remain open for reconsideration. But for the purposes

of this book, we assume that type $\langle e, t \rangle$ entries for vague adjectives like “small” are viable, and we continue to work with the Predicate Modification rule.

We should briefly mention that there are a few other types of apparently nonintersective adjectives, with a behavior rather different from “small”. One group is represented by “former” and “alleged”. Clearly, if John is a former teacher, it does not follow that John is former and John is a teacher. The second conjunct is clearly false in this case, and the first is not even grammatical. The latter fact suggests that “former” does not have a meaning of type $\langle e, t \rangle$, and moreover that it is not, after all, advisable to make a zero modifiee available in the syntax of all VPs of the form “be” + AP. At least, these are the obvious conclusions to draw if we want to predict that “John is former” is plainly uninterpretable. In this respect, “former”, unlike “small”, shows just the distribution that we would expect of an adjective of type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$. But unfortunately its meaning can be shown not to be of this type. The reasoning goes as follows: if $\llbracket \text{former} \rrbracket$ were a function of type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$, then for any two nouns α and β , the following would hold:

(20) If $\llbracket \alpha \rrbracket = \llbracket \beta \rrbracket$, then $\llbracket \text{former } \alpha \rrbracket = \llbracket \text{former } \beta \rrbracket$.

The reason for (20) is that “former α ” (and likewise “former β ”) could only be interpreted by Functional Application: that is, $\llbracket \text{former } \alpha \rrbracket = \llbracket \text{former} \rrbracket(\llbracket \alpha \rrbracket)$. Since $\llbracket \alpha \rrbracket$ is by assumption the same as $\llbracket \beta \rrbracket$, we must have $\llbracket \text{former} \rrbracket(\llbracket \alpha \rrbracket) = \llbracket \text{former} \rrbracket(\llbracket \beta \rrbracket)$. But (20) implies counterintuitive predictions. Suppose Bill’s lovers happen to be exactly the tenants of 13 Green Street. So for any $x \in D$, $\llbracket \text{lover of Bill's} \rrbracket(x) = 1$ iff $\llbracket \text{tenant of 13 Green St} \rrbracket(x) = 1$. By the mathematical definition of a function, this means that $\llbracket \text{lover of Bill's} \rrbracket = \llbracket \text{tenant of 13 Green St} \rrbracket$. With (20), it then follows that, if “John is a former lover of Bill’s” is true, then so is “John is a former tenant of 13 Green Street”. But intuitively, the situation described is entirely compatible with the first of these being true and the second false. So we conclude that $\llbracket \text{former} \rrbracket$ is not of type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$.

Apart from these negative conclusions, our present framework doesn’t enable us to say anything precise. Very roughly, a successful analysis of “former” presupposes a general account of the time parameter in predicates. We must first adapt our semantics to the elementary fact that, for example, “John is a teacher” may be true in 1970 and false in 1980. Our current entries for predicates only make sense if we either disregard any change over time, or tacitly agree that we are considering a certain fixed point in time. The proper treatment of implicit time reference and temporal quantification requires an intensional semantics. An intensional semantics will be introduced in chapter 12, but even there, we will neglect temporal dependencies. Accordingly, we cannot fully answer the question of what the existence of adjectives like “former” implies for the treatment of adjectives and modification in general.

Many other loose ends have been left in this brief introduction to modifiers. The astute reader may have noticed, for instance, that we have said nothing about PPs modifying verbs rather than nouns. A few cautiously chosen examples of this type happen to mean what we predict them to mean: for example, “Julius is sleeping on the couch” does seem to be true just in case Julius is sleeping and is on the couch. (*Exercise:* Show that this is what we predict.) Many other examples that readily come to mind, however, do not have the predicted meanings at all. Consider “John wrote on the blackboard”, “Mary put the book on the table”, or “Max tossed the salad in the bowl”, to name just a few. There are plenty of good grounds here to suspect that our present account of modifiers is very far from how modification really works in natural language. On the other hand, it might turn out that the problems lie elsewhere – for instance, in our simplistic assumptions about verb meaning and VP structure. Indeed, current research overwhelmingly points to the latter conclusion, and many *prima facie* counterexamples emerge as cases of intersective modification after all.¹⁰

4.4 The definite article

We have proposed that common nouns like “cat” denote the characteristic functions of sets of individuals. What does this imply for the semantic analysis of determiners? We will defer the general version of this question to chapter 6. Right here, we will look at only one determiner: namely, the definite article.¹¹

4.4.1 A lexical entry inspired by Frege

The basic intuition about phrases of the form “the NP” is that they denote individuals, just like proper names. Had it not been for Bertrand Russell’s famous claim to the contrary, few people would think otherwise. Frege, for one, thought it obvious: “let us start, e.g., with the expression ‘the capital of the German Empire.’ This obviously takes the place of a proper name, and has as its reference an object.”¹² Hence his practice of referring to definite descriptions as “compound proper names”.

If you read on in that particular passage, Frege imposes a rather odd syntactic analysis on his example. Instead of dividing it into the constituents “the” and “capital of the German Empire”, he splits it up into “the capital of” and “the German Empire”. He then proceeds to analyze “the capital of” as denoting a function from objects to objects (our type $\langle e, e \rangle$). Elsewhere, however, he treats a similar example as follows: